PROJECT DESCRIPTION

Project Title: Revenue Intelligence & Evaluation System (Supabase Backend)

Problem Statement

In structured programs such as competitions, training pipelines, or evaluation systems, it is often difficult to track participant progression, measure performance, and identify where participants drop off.

Without a proper data system, organizers cannot answer key questions such as:

- How many participants progress across stages? - Where do participants drop off?

- Who are the top performers? - Are evaluations consistent and fair?

This limits the ability to make informed decisions and optimize the process.

Solution Overview

To solve this, I designed and implemented a revenue intelligence database system using PostgreSQL (via Supabase).

The system captures the full lifecycle of participant activity, from registration through multi-stage progression to final evaluation.

It transforms raw operational data into actionable insights, enabling performance tracking, participant filtering, and decision-making.

System Design

The database includes the following core entities: - Participants – Categories - Stages (Stage 1 to Stage 5) – Tasks – Submissions – Judges - Evaluations - Stage Unlock System - Audit Logs

A stage progression mechanism was implemented to enforce that participants must complete one stage before moving to the next. This creates a structured flow and enables funnel analysis.

Data Simulation

To reflect real-world behavior:

- 500 participants were generated- 31 categories were created

- Submissions were simulated across multiple stages - Not all participants submitted - Not all submissions met deadlines

This allowed the system to model: - participant drop-off - deadline compliance - stage progression behavior

Key Features

1. Stage Progression Logic

Participants can only move forward if they meet submission requirements.

This enables tracking of conversion rates and drop-off across stages.

2. Category-Based Evaluation

Judges are assigned per category, ensuring that evaluations remain relevant and accurate.

3. Deadline Enforcement

Late submissions are filtered, enabling analysis of submission discipline and time-based behavior.

4. Row Level Security (RLS)

Access is controlled using RLS policies, allowing secure interaction with the system.

5. REST API Integration

The system is exposed via Supabase REST API, making it accessible as a live backed.

Analytical Capabilities

The system supports key analytical queries such as:

- Stage conversion and drop-off analysis

- Deadline compliance tracking

- Top performer identification

- Judge workload distribution

- Category performance comparison - Full funnel analysis (Registered → Submitted → Finalists)

These insights help transform the database into a decision-support system.

Business Impact

This system enables organizers to:

- Identify weak stages and improve participant retention

- Select top performers objectively

- Monitor evaluation fairness and workload

- Track participation trends and engagement

- Make data-driven decisions

Conclusion

This project demonstrates the ability to design a scalable database, simulate realistic behavior, enforce logical constraints, and extract meaningful insights.

It goes beyond data storage by providing intelligence that supports decision-making.